

Subpolynomial dimension reduction in L_p

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Abstract

For every fixed $1 < p < \infty$ and $D > 1$, every n -point subset of a real L_p space embeds into ℓ_p^d with distortion at most D and subpolynomial dimension $d = n^{o(1)}$. The target has the same exponent p , and the embedding need not be linear.

1 Introduction

The Johnson–Lindenstrauss lemma places every n -point subset of a Hilbert space in $O_D(\log n)$ Euclidean dimensions with any prescribed distortion $D > 1$ [11, Lemma 1]. Johnson and Lindenstrauss also asked what analogues of their lemma hold in other Banach spaces [11, Problem 3]. For other L_p spaces, the corresponding finite-metric question must be distinguished both from discretizing an entire linear subspace and from requiring the embedding itself to be linear. We study the coordinate-target version: the points may be repositioned freely, but their images must lie in ℓ_p^d with the same exponent p . For the broader dimension-reduction landscape and the distinction between coordinate targets and arbitrary subspaces, see Naor [15].

All spaces and maps in this paper are real. For $1 < p < \infty$, an integer $n \geq 2$, and $D \geq 1$, let $d_p(n, D)$ be the least integer d such that every n -point subset $\{x_1, \dots, x_n\}$ of any L_p space admits images $y_1, \dots, y_n \in \ell_p^d$ and a number $s > 0$ satisfying

$$s\|x_i - x_j\|_p \leq \|y_i - y_j\|_p \leq Ds\|x_i - x_j\|_p \quad (i, j \in \{1, \dots, n\}). \quad (1.1)$$

The maps are unrestricted; in particular, they need not extend linearly to the ambient space. Lemma 2.1 below shows that the same quantity results if the input sets are restricted to finite coordinate spaces ℓ_p^m .

Theorem 1.1. *Fix $1 < p < \infty$, $p \neq 2$, and put*

$$\gamma(p) = \begin{cases} 2 - p, & 1 < p < 2, \\ 1 - 2/p, & 2 < p < \infty. \end{cases}$$

For every $D > 1$ there is a constant $C_{p,D} < \infty$ such that, for every integer $n \geq 2$,

$$\frac{\log n}{\log(1 + 2D)} \leq d_p(n, D) \leq \exp(C_{p,D}(\log n)^{\gamma(p)}). \quad (1.2)$$

For exact embeddings,

$$\left\lfloor \frac{n-1}{4} \right\rfloor^2 \leq d_p(n, 1) \leq \binom{n}{2} \quad (n \geq 9). \quad (1.3)$$

Consequently,

$$\lim_{n \rightarrow \infty} \frac{\log d_p(n, D)}{\log n} = \begin{cases} 0, & D > 1, \\ 2, & D = 1. \end{cases}$$

The constants in the upper bound depend on the fixed parameters p, D ; no uniformity as p approaches an endpoint or D approaches 1 is asserted. For $p = 2$, the classical Hilbert result gives the stronger bound $d_2(n, D) = O_D(\log n)$ for every fixed $D > 1$. The theorem is existential and does not assert an efficient embedding algorithm.

1.1 Earlier results and the exact-distortion contrast

Ball proved the exact-coordinate upper bound $\binom{n}{2}$ and its quadratic-order sharpness for $1 \leq p < 2$ [1]. Lemma 2.1 recalls the distance-cone and Carathéodory argument for that upper bound. The lower bound in (1.3) uses an antipodal row-and-column configuration. Its disjoint-support test is the classical equality criterion for the p -parallelogram defect [12, Corollary 2.1]; in particular, the argument includes even integers $p > 2$.

Approximate finite-set bounds and linear-subspace discretization have different dimension parameters. Below 2, Schechtman’s finite-set estimates progressed from $O_D(n \log n)$ coordinates to $O_{p,D}(n)$ [22, 23]. The subspace results of Bourgain–Lindenstrauss–Milman and Talagrand instead preserve all vectors in a given r -dimensional subspace by a linear map [3, 25]. Applying such a theorem to the span of the translated data gives $r \leq n - 1$; it does not give the present subpolynomial bound in cardinality. Bourgain’s embedding of arbitrary finite metrics into Hilbert space allows distortion growing with n [2], another different guarantee.

The absence of a linearity requirement is essential: fixed-distortion linear maps on selected L_p configurations can require dimension proportional to their cardinality when $p \neq 2$ [13, Section 3]. At the endpoint $p = 1$, the obstruction of Brinkman–Charikar applies even to nonlinear maps; see also Lee–Naor’s proof [4, 14]. For $p > 2$, Naor–Ren’s recent lower bound [16, Section 1.1, equation (2)] rules out $O_{p,D}(\log n)$ coordinate dimension: along arbitrarily large cardinalities it requires dimension of order at least $(\log n / (\log \log n)^2)^{p/2}$ at fixed distortion. This is compatible with (1.2); neither its upper exponent nor its elementary logarithmic lower bound is asserted to be optimal.

The dependence on cardinality also distinguishes this problem from embedding one infinite space into a fixed finite-dimensional target. A metric space is *doubling* if every ball can be covered by a fixed number of balls of half its radius. A separate construction gives an infinite doubling subset of real Hilbert space with no bi-Lipschitz embedding into any finite-dimensional Euclidean space at any finite distortion [17]. This is compatible with finite-set dimension reduction and is not an input to our proof.

1.2 Method and organization

Our starting point is a bounded distribution of scalar images of the input points. After dividing each increment by the corresponding original distance, we seek nearly equal p th moments for every pair. Lemma 3.1 reduces this simultaneous requirement to a weighted-average condition for each strictly positive probability on the pairs. Crucially, the common moment b and magnitude bound K are uniform over those probabilities. Convex separation produces one distribution working for all pairs; independent sampling then supplies the coordinates of a finite embedding. This sampling step belongs to Maurey’s empirical method, as presented by Pisier [19]. A close distance-matrix

antecedent is Eskenazis’s use of that method for incompressible sets [6, Sections 1.3 and 2.1]; that theorem controls additive errors, with a bound depending on the incompressibility of the data. Here the weighted-moment construction supplies the uniform relative control needed for arbitrary configurations.

The geometric input in Section 4 is a projection onto normalized increments that come from scalar labels on the points. It is an L_2 contraction for a suitable choice of pair weights, and its L_∞ norm is $O(\log n)$. We obtain the latter property by a fixed-point reweighting of the electrical-flow localization estimate of Gurel-Gurevich, Nachmias, and Sachdeva [7, Theorem 1.3]. Their estimate improves the earlier $O(\log^2 n)$ localization bound of Schild–Rao–Srivastava [24, Theorem 1.5, preprint version]. Appendix A presents the heat-kernel and entropy proof from [7, Sections 2–4] in our notation.

After the common homogeneous-increment and signed-Poisson setup, the two ranges require different random increments. Section 6 constructs symmetric p -stable marginals, clips them, and projects back to consistent increments. Stable projections and truncated heavy-tail moments have earlier geometric uses, including Indyk’s work [10]; the required marginal construction and tail bounds are proved here. For $p > 2$, Section 7 uses overlapping magnitude truncations followed by signed Poisson sampling. Their overlap enlarges the input energy relative to the projection error. The moment comparison is of Rosenthal type [21, 18]; Appendix B proves the specific leading coefficient $1 + \epsilon$ needed for distortion arbitrarily close to one. The final main-text section inserts the resulting b, K into the sampling estimate and proves the packing and exact embedding lower bounds.

We use finite-dimensional convexity, Brouwer’s fixed-point theorem, and standard probability; the specialized electrical-localization and Poisson-moment arguments are supplied in the appendices.

Throughout, logarithms are natural. Constants $C, c > 0$ are absolute; subscripts indicate the parameters on which a constant may depend. Their values may change from line to line.

2 Finite coordinates and normalized increments

Lemma 2.1 (Exact discretization). *Every n -point subset of a real L_p space embeds isometrically into $\ell_p^{\binom{n}{2}}$. This holds for every $1 \leq p < \infty$.*

Proof. Let the points be $x_1, \dots, x_n \in L_p(\mu)$, and let E be the set of unordered pairs of distinct indices. Write $N = |E|$. Choose measurable representatives, finite outside a common null set, and put

$$q(t) = (|x_i(t) - x_j(t)|^p)_{ij \in E}, \quad s(t) = \sum_{e \in E} q_e(t), \quad S = \int s \, d\mu.$$

The points are distinct, so $0 < S < \infty$. Consider the compact set

$$\mathcal{K} = \left\{ (|z_i - z_j|^p)_{ij \in E} : z \in \mathbb{R}^n, \, z_n = 0, \, \sum_{ij \in E} |z_i - z_j|^p = 1 \right\} \subset \mathbb{R}^E.$$

Compactness follows because its defining set of labels is closed and $|z_i| \leq 1$ for every i . For $s(t) > 0$, the vector $q(t)/s(t)$ belongs to $\mathcal{K}$. Hence

$$\frac{1}{S} \int q(t) \, d\mu(t) = \int_{\{s>0\}} \frac{q(t)}{s(t)} \frac{s(t)}{S} \, d\mu(t) \in \text{conv}(\mathcal{K}).$$

Here a barycenter belongs to the convex hull because that convex hull is compact; equivalently, this follows by finite-dimensional separation [20, Theorems 11.3 and 17.2]. The set $\mathcal{K}$ lies in the affine hyperplane whose coordinates sum to one. By Carathéodory's theorem [20, Theorem 17.1], the displayed vector is a convex combination of at most N points of $\mathcal{K}$. Choose corresponding labels $z^{(a)}$ and coefficients $\theta_a \geq 0$ with $\sum_a \theta_a = 1$. Then

$$y_i = ((S\theta_a)^{1/p} z_i^{(a)})_a$$

satisfies $\|y_i - y_j\|_p^p = \|x_i - x_j\|_p^p$ for every pair. $\square$

For the upper bound in (1.2), we may therefore fix distinct points $x_1, \dots, x_n \in \ell_p^m$ for some finite m . Give the complete graph on $\{1, \dots, n\}$ an arbitrary orientation, and use $e = ij$ to denote an edge with this ordering. Set

$$\delta_e = \|x_i - x_j\|_p > 0, \quad V = \left\{ \left(\frac{z_i - z_j}{\delta_e} \right)_{e=ij \in E} : z \in \mathbb{R}^n \right\} \subset \mathbb{R}^E.$$

We call elements of V *normalized gradients*. Each such vector determines scalar labels z_i uniquely after imposing $z_n = 0$; this choice is linear in the vector. Thus a collection of normalized gradients can always be used as the coordinates of one map on the given points.

For a strictly positive probability vector $\sigma = (\sigma_e)_{e \in E}$, write

$$\|f\|_{2,\sigma}^2 = \sum_{e \in E} \sigma_e |f_e|^2.$$

We first state the moment criterion that our random gradients must satisfy.

3 From weighted moments to an embedding

A random normalized gradient gives candidate scalar coordinates. We seek one bounded distribution whose p th moments are nearly equal on every edge. The next lemma shows that it is enough to meet each weighted test separately, even though the distribution may change with the test probability. The common moment and magnitude bound must remain fixed.

The common moment b below is allowed to grow: multiplying all distances by the same factor does not change distortion. This freedom will let us absorb an additive error in the moment estimates.

Lemma 3.1 (Weighted moment criterion). *Let $0 < \eta < 1/3$ and $b, K \geq 1$. Suppose that, for every strictly positive probability vector λ on E , there is a random normalized gradient $F \in V$ such that*

$$\|F\|_\infty \leq K \quad \text{almost surely,} \quad \sum_{e \in E} \lambda_e |\mathbb{E}|F_e|^p - b| \leq \eta b. \quad (3.1)$$

The numbers b and K are the same for all λ . Then the points admit an embedding into ℓ_p^d with distortion at most

$$\left(\frac{1 + 3\eta}{1 - 3\eta} \right)^{1/p},$$

where

$$d \leq \left\lceil C \max \left\{ 1, \frac{K^{2p}}{\eta^2 b^2} \right\} \log(2n) \right\rceil \quad (3.2)$$

for an absolute constant C .

Proof. Let $\mathcal{C}$ be the convex hull of the compact set

$$\left\{ (|f_e|^p)_{e \in E} : f \in V, \|f\|_\infty \leq K \right\}.$$

It is compact and consists exactly of the moment vectors of distributions of such gradients; finite mixtures suffice by Carathéodory's theorem. We claim that $\mathcal{C}$ meets the open box

$$\mathcal{B} = \{m \in \mathbb{R}^E : |m_e - b| < 2\eta b \text{ for all } e\}.$$

Otherwise, finite-dimensional separation [20, Theorem 11.3] gives a vector $a \neq 0$ such that, after normalizing $\sum_e |a_e| = 1$,

$$\sum_e a_e(m_e - b) \geq 2\eta b \quad (m \in \mathcal{C}).$$

Apply the hypothesis with $\lambda_e = \frac{3}{4}|a_e| + \frac{1}{4|E|}$. Its moment vector $m \in \mathcal{C}$ satisfies

$$\sum_e a_e(m_e - b) \leq \sum_e |a_e| |m_e - b| \leq \frac{4}{3} \sum_e \lambda_e |m_e - b| \leq \frac{4}{3} \eta b,$$

a contradiction. Thus there is one distribution of gradients, still bounded by K , for which every expected p th moment differs from b by less than $2\eta b$.

For completeness, we prove a Hoeffding-type bounded-sum estimate for the sampling step, with constants weaker than the classical bound in [9, Theorem 2, p. 16, equation (2.6)]. If $0 \leq X \leq K^p$ and X' is an independent copy, Jensen's inequality and symmetry give

$$\mathbb{E} e^{t(X - \mathbb{E}X)} \leq \mathbb{E} e^{t(X - X')} = \mathbb{E} \cosh(t(X - X')) \leq e^{t^2 K^{2p}/2},$$

using $\cosh u \leq e^{u^2/2}$. Take d independent samples $F^{(1)}, \dots, F^{(d)}$ from the preceding distribution. Apply this bound to $X = |F_e|^p$, multiply moment generating functions, and use Markov's inequality with $t = \eta b / K^{2p}$. Applying the same argument to the lower tail gives

$$\mathbb{P} \left(\left| \frac{1}{d} \sum_{a=1}^d |F_e^{(a)}|^p - \mathbb{E}|F_e|^p \right| > \eta b \right) \leq 2 \exp \left(-\frac{d\eta^2 b^2}{2K^{2p}} \right).$$

For the value in (3.2) with C sufficiently large, a union bound over $|E| < n^2/2$ edges is less than one. Choose an outcome for which all empirical moments belong to $[(1 - 3\eta)b, (1 + 3\eta)b]$. Realize each $F^{(a)}$ by scalar labels $z_i^{(a)}$, and set

$$y_i = d^{-1/p} (z_i^{(1)}, \dots, z_i^{(d)}).$$

Then

$$\frac{\|y_i - y_j\|_p^p}{\delta_{ij}^p} = \frac{1}{d} \sum_{a=1}^d |F_{ij}^{(a)}|^p.$$

This proves the assertion, with scale $s = ((1 - 3\eta)b)^{1/p}$. $\square$

4 A projection with controlled row sums

The projection estimate follows from the electrical-flow localization theorem of Gurel-Gurevich, Nachmias, and Sachdeva [7, Theorem 1.3]. For a matrix T , write $|T|$ for its entrywise absolute value.

Lemma 4.1 (Electrical-flow localization). *Let G be a connected loopless graph on $n \geq 2$ vertices, let B be an oriented edge-by-vertex incidence matrix, and let $C = \text{diag}(c_e)$ with $c_e > 0$. If Q is the Euclidean orthogonal projection onto $\text{im}(C^{1/2}B)$, then*

$$\| |Q| \|_{2 \rightarrow 2} \leq 2 \log n. \quad (4.1)$$

The proof is given in Appendix A.

We now apply this estimate to the normalized-gradient space V . Set

$$H = 4 \log(2n). \quad (4.2)$$

Lemma 4.2 (Weighted projection). *For every strictly positive probability vector λ on E , there is a probability vector σ with $\sigma_e \geq \lambda_e/2$ such that the orthogonal projection P onto V in $\ell_2(\sigma)$ satisfies*

$$\max_{e \in E} \sum_{f \in E} |P_{ef}| \leq H. \quad (4.3)$$

In particular, $\|P\|_{\infty \rightarrow \infty} \leq H$, and both P and $I - P$ are contractions on $\ell_2(\sigma)$.

Proof. Let B be the incidence matrix of the oriented complete graph fixed above. For a strictly positive probability vector σ , denote the weighted orthogonal projection by $P(\sigma)$, and put

$$D_\sigma = \text{diag}(\sqrt{\sigma_e}), \quad C_\sigma = \text{diag}(\sigma_e/\delta_e^2).$$

Then $Q(\sigma) = D_\sigma P(\sigma) D_\sigma^{-1}$ is the Euclidean orthogonal projection onto $\text{im}(C_\sigma^{1/2}B)$. Define $R_e(\sigma) = \sum_f |P(\sigma)_{ef}|$. Since $\|\sqrt{\sigma}\|_2 = 1$, Lemma 4.1 gives

$$\sum_e \sigma_e R_e(\sigma)^2 = \| |Q(\sigma)| \sqrt{\sigma} \|_2^2 \leq (2 \log n)^2. \quad (4.4)$$

To control every row, we choose weights that allocate more mass to rows with large absolute sums. Consider the compact convex set

$$\Sigma = \left\{ \sigma : \sum_e \sigma_e = 1, \sigma_e \geq \lambda_e/2 \right\}.$$

The weighted projection, and hence every R_e , depends continuously on $\sigma \in \Sigma$: in a fixed basis of V , its formula involves the inverse of a positive definite Gram matrix. Set

$$u_e(\sigma) = \frac{\sigma_e R_e(\sigma)^2}{2(2 \log n)^2}, \quad \Phi_e(\sigma) = u_e(\sigma) + \left(1 - \sum_f u_f(\sigma) \right) \lambda_e.$$

By (4.4), $\sum_e u_e(\sigma) \leq 1/2$. Thus Φ is a continuous self-map of Σ , and Brouwer's theorem [8, Corollary 2.15] provides a fixed point. At that fixed point $u_e(\sigma) \leq \sigma_e$; since $\sigma_e > 0$, this gives

$$R_e(\sigma) \leq 2\sqrt{2} \log n \leq H.$$

The remaining assertions follow from the row-sum formula for the ℓ_∞ operator norm and orthogonality in $\ell_2(\sigma)$. $\square$

For the rest of the upper-bound argument, fix $D > 1$ and choose $0 < \eta < 1/6$ so that

$$\left(\frac{1+3\eta}{1-3\eta}\right)^{1/p} \leq D. \quad (4.5)$$

For each strictly positive test probability λ , we will use σ and P from Lemma 4.2. It suffices to construct F with weighted error at most $\eta b/2$ under σ , since $\lambda \leq 2\sigma$ coordinatewise. The parameters b and K will depend only on p, η, n .

5 Homogeneous increments and signed Poisson sampling

The following measure gives every normalized increment the same magnitude distribution. On $\Omega = (0, \infty) \times \{1, \dots, m\}$, set

$$d\nu(r, t) = pr^{-p-1} dr \quad \text{on each copy indexed by } t, \quad g_e(t) = \frac{x_i(t) - x_j(t)}{\delta_e}, \quad v_e(r, t) = rg_e(t).$$

For every (r, t) , the vector $v(r, t)$ belongs to V . Since $\sum_{t=1}^m |g_e(t)|^p = 1$, direct integration gives

$$\nu(|v_e| > a) = a^{-p} \quad (a > 0). \quad (5.1)$$

More generally, for every nonnegative measurable function $\phi : (0, \infty) \rightarrow [0, \infty]$,

$$\int_{\{v_e \neq 0\}} \phi(|v_e|) d\nu = p \int_0^\infty \phi(u) u^{-p-1} du. \quad (5.2)$$

In particular, $\nu(\max_e |v_e| > a) \leq |E|a^{-p}$. The measure ν is not a probability measure. The second moment of the small jumps is finite precisely when $p < 2$; for $p > 2$, the useful finite second moment is instead in the large jumps. This leads to two different uses of the same homogeneous increments.

We record the elementary sampling facts used in both ranges. On a measurable region of finite mass M , *signed Poisson sampling* of a vector function h means

$$W = \sum_{j=1}^N \varepsilon_j h(\omega_j),$$

where N has Poisson law of mean M , the ω_j are independent with law ν/M restricted to the region, and the ε_j are independent uniform signs. All these choices are independent. For $M = 0$ the sum is zero. Sampling on disjoint regions is independent: indeed, subdividing a Poisson number of independent samples gives independent Poisson counts, as is seen from their joint probability generating function. In particular, sampling a V -valued function produces a vector in V .

Lemma 5.1 (Signed Poisson formulas). *Let h be a bounded real function supported on a region of finite ν -measure, and let S be its signed Poisson sum. Then*

$$\mathbb{E}e^{itS} = \exp\left(\int (\cos(th) - 1) d\nu\right), \quad (5.3)$$

$$\mathbb{E}e^{tS} = \exp\left(\int (\cosh(th) - 1) d\nu\right), \quad (5.4)$$

$$\mathbb{E}S = 0, \quad \mathbb{E}S^2 = \int h^2 d\nu. \quad (5.5)$$

The characteristic-function identity also holds for unbounded h on a finite-measure region.

Proof. Conditioning on the Poisson count proves the first two formulas: the expectation for one signed sample is respectively the average of $\cos(th)$ or $\cosh(th)$, and the Poisson generating function is $\mathbb{E}z^N = \exp(M(z-1))$. The variance formula follows by conditioning as well. $\square$

6 The range $1 < p < 2$

Fix $1 < p < 2$. We construct the random normalized gradient required by Lemma 3.1, with η as chosen in (4.5).

6.1 A common heavy-tailed marginal

We first construct a random vector $Y \in V$ by symmetric Poisson sampling of the vector v against the intensity ν . On the region $\{\|v\|_\infty > 1\}$, this is an ordinary signed compound Poisson sum: the region has measure at most $|E|$ by (5.1). For each $j \geq 0$, independently sample the region

$$A_j = \{2^{-j-1} < \|v\|_\infty \leq 2^{-j}\},$$

and denote its signed compound Poisson sum by $Y^{(j)}$. Each A_j has finite measure, at most $|E|2^{p(j+1)}$. These are centered independent vectors. For every edge e ,

$$\sum_{j \geq 0} \mathbb{E}|Y_e^{(j)}|^2 = \int_{0 < \|v\|_\infty \leq 1} |v_e|^2 d\nu \leq \int_{0 < |v_e| \leq 1} |v_e|^2 d\nu = \frac{p}{2-p}.$$

Consequently $\sum_{j \geq 0} Y^{(j)}$ converges in L_2 with values in the finite dimensional space $\mathbb{R}^E$. Adding the sum on $\{\|v\|_\infty > 1\}$ defines Y . Every partial sum belongs to V ; since V is closed, $Y \in V$ almost surely.

The compound Poisson formula and convergence in probability give

$$\mathbb{E}e^{i\theta Y_e} = \exp\left(\int (\cos(\theta v_e) - 1) d\nu\right) = \exp\left(p \int_0^\infty (\cos(\theta u) - 1) u^{-p-1} du\right). \quad (6.1)$$

The integrals converge absolutely: near zero the integrand is $O_\theta(u^{1-p})$, and at infinity it is $O(u^{-p-1})$. The second equality follows from (5.2) and the evenness of cosine. In particular, uniqueness of characteristic functions [5, Theorem 3.3.11] shows that all the Y_e have the same distribution, depending only on p .

For later use we prove directly that

$$c_p a^{-p} \leq \mathbb{P}(|Y_e| > a) \leq C_p a^{-p} \quad (a \geq 1). \quad (6.2)$$

Fix e and $a \geq 1$. By Poisson splitting, or by the characteristic function (6.1), Y_e has the distribution of an independent sum $B_a + S_a$. Here B_a is the signed compound Poisson sum on $\{|v_e| > a\}$, whose number N_a of jumps is Poisson with mean a^{-p} , while S_a is the L_2 limit of the small-jump sums on $\{0 < |v_e| \leq a\}$. Thus

$$\mathbb{E}S_a = 0, \quad \mathbb{E}|S_a|^2 = \int_{0 < |v_e| \leq a} |v_e|^2 d\nu = \frac{p}{2-p} a^{2-p}.$$

Chebyshev's inequality gives

$$\mathbb{P}(|Y_e| > a) \leq \mathbb{P}(N_a \geq 1) + \mathbb{P}(|S_a| > a) \leq \left(1 + \frac{p}{2-p}\right) a^{-p}.$$

For the reverse inequality, condition on $N_a = 1$, the magnitude $u > a$ of its jump, and $S_a = s$. At least one of $|s + u|$ and $|s - u|$ exceeds a , and the jump's two signs are equally likely. Therefore

$$\mathbb{P}(|Y_e| > a) \geq \frac{1}{2}\mathbb{P}(N_a = 1) = \frac{1}{2}a^{-p}e^{-a^{-p}} \geq \frac{1}{2e}a^{-p}.$$

This proves (6.2).

6.2 Clipping and projection

Let λ be a strictly positive probability on E , and choose σ and P from Lemma 4.2. Thus $\sigma \geq \lambda/2$, and P has absolute row sums at most H . For $T > 1$, let $[Y]_T$ denote coordinatewise clipping to $[-T, T]$, and set

$$F = P[Y]_T.$$

Then $F \in V$ and $\|F\|_\infty \leq HT$ almost surely. Since $Y = PY$ and $p > 1$, integration of (6.2) yields

$$\begin{aligned} \mathbb{E}|Y_e - F_e| &\leq \sum_{f \in E} |P_{ef}| \mathbb{E}(|Y_f| - T)_+ \\ &\leq C_p HT^{1-p}. \end{aligned} \tag{6.3}$$

Orthogonal contraction, together with the same tail bound, gives

$$\begin{aligned} \sum_e \sigma_e \mathbb{E}|F_e|^2 &\leq \sum_e \sigma_e \mathbb{E} \min\{|Y_e|, T\}^2 \\ &\leq C_p T^{2-p}. \end{aligned} \tag{6.4}$$

We compare moments below a second level a . The Lipschitz comparison will cost $H(a/T)^{p-1}$, while the quadratic estimate for the remaining tail will cost $(T/a)^{2-p}$. Balancing these gives $a = T/H$. Suppose $a \geq 2$, and define the common capped moment

$$b = \mathbb{E} \min\{|Y_e|, a\}^p.$$

The common marginal law makes b independent of e and λ . Moreover, the layer-cake formula and (6.2) imply

$$b = p \int_0^a t^{p-1} \mathbb{P}(|Y_e| > t) dt \geq c_p \log a. \tag{6.5}$$

The function $h_a(t) = \min\{|t|, a\}^p$ is pa^{p-1} -Lipschitz. Hence (6.3) implies, for every e ,

$$|\mathbb{E}h_a(F_e) - b| \leq C_p a^{p-1} HT^{1-p} = C_p H^{2-p}.$$

The part above this cap is controlled in weighted average. Since $p < 2$, for every real t we have

$$0 \leq |t|^p - h_a(t) \leq a^{p-2}|t|^2.$$

Using (6.4), we obtain

$$\sum_e \sigma_e \mathbb{E}(|F_e|^p - h_a(F_e)) \leq C_p a^{p-2} T^{2-p} = C_p H^{2-p}.$$

Combining the two estimates proves

$$\sum_e \sigma_e |\mathbb{E}|F_e|^p - b| \leq C_p H^{2-p}. \tag{6.6}$$

Choose

$$T = \exp(A_{p,\eta} H^{2-p}), \quad K = HT,$$

where $A_{p,\eta}$ is sufficiently large. Since $\log H = o(H^{2-p})$, for all sufficiently large n we have $a \geq 2$ and

$$b \geq c_p \log(T/H) \geq \frac{c_p A_{p,\eta}}{2} H^{2-p} \geq 1.$$

By (6.6) and $\lambda \leq 2\sigma$, increasing $A_{p,\eta}$ if necessary ensures

$$\sum_e \lambda_e |\mathbb{E}|F_e|^p - b| \leq \eta b.$$

Both b and K are independent of the test probability λ . Finally,

$$\log K = \log H + A_{p,\eta} H^{2-p} = O_{p,\eta}((\log n)^{2-p}).$$

These are the parameters required by Lemma 3.1 in this range.

7 The range $2 < p < \infty$

Fix $p > 2$, with η as in (4.5). For a strictly positive test probability λ , obtain $\sigma \geq \lambda/2$ and P from Lemma 4.2. Write

$$\gamma = 1 - \frac{2}{p}, \quad \rho = \sigma \otimes \nu.$$

All function norms below are taken with respect to ρ , unless another measure is indicated. The projection acts only in the edge variable.

7.1 A ramp of truncations

The overlapping truncations below create a triangular multiplicity profile of height ℓ . We will show that its p -energy is of order ℓ^{p+1} , whereas the p th power of the projection error is at most $C_p H^{p-2} \ell$. This gain determines the choice of ℓ .

For $a > 0$, let $v_{\leq a}$ denote the vector obtained by replacing v_e by $v_e \mathbf{1}_{\{|v_e| \leq a\}}$ at each edge; put $v_{>a} = v - v_{\leq a}$. For an integer $\ell \geq 1$, to be chosen below, set

$$u = \sum_{k=0}^{\ell-1} (v_{\leq 2^{k+\ell}} - v_{\leq 2^k}), \quad Z = Pu. \tag{7.1}$$

Thus u_e is v_e multiplied by the number of intervals $(2^k, 2^{k+\ell}]$, $0 \leq k < \ell$, containing $|v_e|$. The successive nonzero multiplicities on the dyadic magnitude intervals are

$$1, 2, \dots, \ell-1, \ell, \ell-1, \dots, 2, 1.$$

The magnitude-density identity (5.2) therefore gives the same p -energy at every edge:

$$b := \int |u_e|^p d\nu = p \log 2 \left(\ell^p + 2 \sum_{j=1}^{\ell-1} j^p \right), \quad c_p \ell^{p+1} \leq b \leq C_p \ell^{p+1}. \tag{7.2}$$

In particular, $b \geq p \log 2 > 1$, and b is independent of both the input points and λ . Figure 1 illustrates this energy calculation for four overlapping intervals.

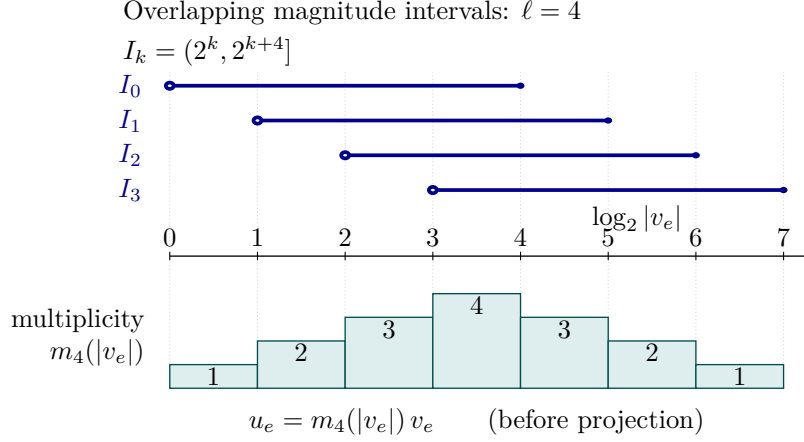


Figure 1: The overlapping truncations for $\ell = 4$. The intervals $I_k = (2^k, 2^{k+4}]$, $0 \leq k < 4$, give multiplicities 1, 2, 3, 4, 3, 2, 1 on $(2^j, 2^{j+1}]$, $j = 0, \dots, 6$, and zero outside $(1, 128]$. Thus $u_e = m_4(|v_e|)v_e$ before projection. By the magnitude-density identity, a bin of multiplicity m contributes $p \log 2 m^p$ to $\int |u_e|^p d\nu$.

The same identity gives

$$\int |v_e|^2 \mathbf{1}_{\{|v_e| > a\}} d\nu = \frac{p}{p-2} a^{2-p}.$$

The triangle inequality in $L_2(\nu)$, applied to (7.1), consequently yields

$$\left(\int |u_e|^2 d\nu \right)^{1/2} \leq \left(\frac{p}{p-2} \right)^{1/2} \sum_{k=0}^{\ell-1} 2^{(1-p/2)k} \leq C_p.$$

Using the contraction of P in $\ell_2(\sigma)$ and its row-sum bound, we obtain

$$\|Z\|_{L_2(\rho)}^2 \leq C_p, \quad \sup_e \int |Z_e|^2 d\nu \leq C_p H^2, \quad \sup_{e,r,t} |Z_e(r,t)| \leq H 2^{2\ell}. \quad (7.3)$$

In particular, the coordinatewise quadratic estimate follows from

$$\left(\int |Z_e|^2 d\nu \right)^{1/2} \leq \sum_f |P_{ef}| \left(\int |u_f|^2 d\nu \right)^{1/2} \leq C_p H.$$

For the supremum bound in (7.3), each summand defining u_e has magnitude at most $2^{k+\ell}$, and their sum is less than $2^{2\ell}$.

7.2 The projection error

We claim that

$$\|Z - u\|_{L_p(\rho)}^p \leq C_p H^{p-2} (\ell + 1). \quad (7.4)$$

For $0 \leq k < 2\ell$, define the pointwise vector function

$$A_k = (P - I)v_{\leq 2^k}.$$

Although $v_{\leq 2^k}$ need not belong to $L_2(\rho)$, the identity $Pv = v$ holds pointwise and gives $A_k = -(P - I)v_{>2^k}$. Thus A_k does belong to $L_2(\rho)$, and

$$\|A_k\|_\infty \leq 2H2^k, \quad \|A_k\|_{L_2(\rho)} \leq C_p 2^{(1-p/2)k}. \quad (7.5)$$

Here the second inequality uses the $\ell_2(\sigma)$ contraction of $I - P$ and the preceding tail integral. Moreover,

$$Z - u = \sum_{k=\ell}^{2\ell-1} A_k - \sum_{k=0}^{\ell-1} A_k.$$

Summing the two bounds in (7.5) separately shows that this signed sum has $L_2(\rho)$ norm at most C_p and supremum norm at most $CH2^{2\ell}$.

For a level $\tau \geq 16H$, split the sum at $2^k \leq \tau/(16H)$. The sum of the supremum norms of these smaller terms is at most $\tau/2$. By geometric decay, the remaining terms have total $L_2(\rho)$ norm at most $C_p(\tau/H)^{1-p/2}$. Chebyshev's inequality, together with the global L_2 bound for smaller levels, gives

$$\rho(|Z - u| > \tau) \leq \begin{cases} C_p \tau^{-2}, & 0 < \tau < 16H, \\ C_p H^{p-2} \tau^{-p}, & \tau \geq 16H. \end{cases}$$

This tail vanishes for $\tau > CH2^{2\ell}$. Integrating it against $p\tau^{p-1} d\tau$ proves (7.4): the smaller levels contribute $C_p H^{p-2}$, and the larger levels contribute at most $C_p H^{p-2}(\ell + 1)$.

Since $\|u\|_{L_p(\rho)}^p = b$, the pointwise power-difference inequality and Hölder's inequality imply

$$\sum_e \sigma_e \left| \int |Z_e|^p d\nu - b \right| \leq p \|Z - u\|_{L_p(\rho)} (\|u\|_{L_p(\rho)} + \|Z - u\|_{L_p(\rho)})^{p-1}. \quad (7.6)$$

By (7.2) and (7.4),

$$\frac{\|Z - u\|_{L_p(\rho)}}{b^{1/p}} \leq C_p \frac{H^\gamma}{\ell}. \quad (7.7)$$

Choose

$$\ell = \lceil LH^\gamma \rceil, \quad (7.8)$$

where $L \geq 1$ depends only on p, η and will be enlarged once more below. First choose it large enough that (7.6) is at most $\eta b/8$.

7.3 Poisson sampling

We need the leading coefficient in the next estimate to be arbitrarily close to one, since the desired distortion D may be arbitrarily close to one.

Lemma 7.1 (Nearly additive Poisson moments). *Let h be a bounded real function supported on a region of finite ν -measure, and let S be its signed Poisson sum. For $q > 2$ and $\epsilon > 0$,*

$$\int |h|^q d\nu \leq \mathbb{E}|S|^q \leq (1 + \epsilon) \int |h|^q d\nu + C_{q,\epsilon} \left(\int h^2 d\nu \right)^{q/2}. \quad (7.9)$$

A proof using independent symmetric sums is given in Appendix B.

The function Z vanishes outside

$$S = \{(r, t) : \max_e |v_e(r, t)| > 1\}, \quad \nu(S) \leq |E|,$$

because u vanishes there. Hence we may take the signed Poisson sum W of Z on this finite-measure region. Every summand is in V , so $W \in V$ almost surely.

Write $m_e = \int |Z_e|^p d\nu$ and $a_e = \int |Z_e|^2 d\nu$. Lemma 7.1 gives, for every $\epsilon > 0$,

$$0 \leq \mathbb{E}|W_e|^p - m_e \leq \epsilon m_e + C_{p,\epsilon} a_e^{p/2}.$$

The two quadratic estimates in (7.3) imply

$$\sum_e \sigma_e a_e^{p/2} \leq \left(\max_e a_e \right)^{p/2-1} \sum_e \sigma_e a_e \leq C_p H^{p-2}. \quad (7.10)$$

Set $\epsilon = \eta/32$. The projection estimate gives $\sum_e \sigma_e m_e \leq (1 + \eta/8)b$, so the averaged ϵm_e term is at most $\eta b/16$. Further,

$$\frac{H^{p-2}}{b} \leq C_p \frac{H^{p-2}}{\ell^{p+1}} \leq C_p L^{-(p+1)} H^{-\gamma} \leq C_p L^{-(p+1)}.$$

Enlarge L , still depending only on p, η , until the contribution of (7.10), including its constant $C_{p,\epsilon}$, is at most $\eta b/16$. We have proved

$$\sum_e \sigma_e |\mathbb{E}|W_e|^p - b| \leq \frac{\eta b}{8} + \frac{\eta b}{16} + \frac{\eta b}{16} = \frac{\eta b}{4}. \quad (7.11)$$

7.4 A bounded random gradient

Choose a constant C'_p so that

$$J = C'_p H 2^{2\ell} \geq 1$$

dominates both $\sup_{e,r,t} |Z_e(r,t)|$ and $\sup_e a_e^{1/2}$. The exponential-moment formula for signed Poisson sums gives

$$\mathbb{E} \exp(\pm W_e/J) = \exp \left(\int (\cosh(Z_e/J) - 1) d\nu \right) \leq e^{C_0},$$

where C_0 is absolute: use $\cosh s - 1 \leq C_0 s^2$ for $|s| \leq 1$ and $a_e \leq J^2$. Consequently, for $z \geq 0$,

$$\mathbb{P}(|W_e| > Jz) \leq 2e^{C_0-z}, \quad \mathbb{E}|W_e|^{2p} \leq C_p J^{2p}. \quad (7.12)$$

Set

$$K = 8J \log(2n), \quad \mathcal{B} = \{\max_e |W_e| > K\}, \quad F = W \mathbf{1}_{\mathcal{B}^c}.$$

Discarding the entire vector on a bad event preserves membership in V , so $F \in V$ and $\|F\|_\infty \leq K$. By the union bound and (7.12),

$$\mathbb{P}(\mathcal{B}) \leq 2|E|e^{C_0}(2n)^{-8} \leq Cn^{-6}.$$

For every edge, Cauchy–Schwarz therefore gives

$$0 \leq \mathbb{E}|W_e|^p - \mathbb{E}|F_e|^p \leq (\mathbb{E}|W_e|^{2p})^{1/2} \mathbb{P}(\mathcal{B})^{1/2} \leq C_p J^p n^{-3}. \quad (7.13)$$

By (7.8),

$$\log J = O_{p,\eta}(H^\gamma) = o(\log n).$$

Thus the last bound in (7.13) tends to zero uniformly over the input points and λ . Since $b \geq 1$, it is at most $\eta b/4$ for all sufficiently large n , with the threshold depending only on p, η . Combining this with (7.11) and $\lambda \leq 2\sigma$ yields

$$\sum_e \lambda_e |\mathbb{E}|F_e|^p - b| \leq 2 \sum_e \sigma_e |\mathbb{E}|F_e|^p - b| \leq \eta b.$$

The parameters $b \geq 1$ and $K \geq 1$ are independent of λ and of the input points, as required by Lemma 3.1. Finally,

$$\log K = O_{p,\eta}(H^\gamma) = O_{p,\eta}((\log n)^{1-2/p}).$$

These are the parameters required by Lemma 3.1 in this range.

8 Dimension bounds and exact embeddings

Table 1 summarizes the common parameters supplied by the two constructions. The target moment and magnitude bound are fixed before the test probability λ varies; the projection and the joint law of the random gradient may depend on that probability.

Range	Common moment b	Magnitude bound K
$1 < p < 2$	$\mathbb{E} \min\{ Y_e , T/H\}^p$	HT
$p > 2$	$p \log 2(\ell^p + 2 \sum_{1 \leq j < \ell} j^p)$	$8J \log(2n)$

Table 1: The weighted-moment contract for sufficiently large n . Here $T = \exp(A_{p,\eta} H^{2-p})$, $\ell = \lceil LH^{1-2/p} \rceil$ in their respective ranges, and $J = C'_p H 2^{2\ell}$. All these choices are independent of the input points and pair weights.

8.1 Completion of the upper bound

In Sections 6 and 7, the constructed parameters satisfy $b, K \geq 1$, the weighted moment condition (3.1), and

$$\log K \leq C_{p,\eta}(\log n)^{\gamma(p)}$$

for all sufficiently large n , with a threshold depending only on p, η . Lemma 3.1 and (4.5) give an embedding of distortion at most D . Since $b \geq 1$, its dimension satisfies

$$\log d \leq C_{p,\eta} + 2p \log K + \log \log(2n) \leq C_{p,D}(\log n)^{\gamma(p)}.$$

This proves the upper bound in (1.2) for sufficiently large n . For the remaining finitely many values of n , enlarge $C_{p,D}$ and use Lemma 2.1.

8.2 The packing bound

The coordinate unit vectors in ℓ_p^n form an equilateral set. After translating and scaling any distortion- D image in ℓ_p^d , its points are 1-separated and lie in the ball of radius D about the first point. Their open balls of radius $1/2$ are disjoint and lie in the ball of radius $D + 1/2$ about that point. Comparing d -dimensional Lebesgue volumes gives

$$n(1/2)^d \leq (D + 1/2)^d, \quad \text{hence} \quad n \leq (1 + 2D)^d.$$

This proves the lower bound in (1.2).

8.3 A quadratic obstruction to exact embeddings

For real a, b , define

$$\Delta_p(a, b) = |a + b|^p + |a - b|^p - 2|a|^p - 2|b|^p.$$

If $ab = 0$, this is zero. If $ab \neq 0$, it has strictly the sign of $p - 2$. To verify this, apply convexity for $p > 2$, or concavity for $p < 2$, to the two numbers $(a + b)^2$ and $(a - b)^2$ raised to the power $p/2$. Their average is $a^2 + b^2$. Next use the strict inequality

$$(a^2 + b^2)^{p/2} \begin{cases} > |a|^p + |b|^p, & p > 2, \\ < |a|^p + |b|^p, & p < 2. \end{cases}$$

Thus, for $u, v \in \ell_p^d$ and $p \neq 2$,

$$\|u + v\|_p^p + \|u - v\|_p^p = 2\|u\|_p^p + 2\|v\|_p^p \iff \text{supp}(u) \cap \text{supp}(v) = \emptyset. \quad (8.1)$$

Indeed, the difference of the two sides is $\sum_{a=1}^d \Delta_p(u_a, v_a)$, whose nonzero summands all have the same sign.

Fix $k \geq 2$. Index the coordinates of $\ell_p^{k^2}$ by $\{1, \dots, k\}^2$. Let r_i be the indicator of row i , and c_j the indicator of column j . Consider the $4k + 1$ distinct points

$$\mathcal{X}_k = \{0\} \cup \{r_i, -r_i : 1 \leq i \leq k\} \cup \{c_j, -c_j : 1 \leq j \leq k\}.$$

Suppose that $\mathcal{X}_k$ embeds with distortion one into ℓ_p^d . Translate and rescale to obtain an isometry f with $f(0) = 0$. Strict convexity of ℓ_p^d implies

$$f(-x) = -f(x) \quad (x \in \mathcal{X}_k). \quad (8.2)$$

In fact, $\|f(x) - f(-x)\|_p = 2\|x\|_p$ and both $\|f(x)\|_p$ and $\|f(-x)\|_p$ equal $\|x\|_p$, so equality in the triangle inequality forces the two vectors $f(x)$ and $-f(-x)$ to coincide.

By (8.2), the expression in (8.1) is determined by distances among the given points and is preserved by f . Distinct rows have disjoint supports, so the supports of $f(r_1), \dots, f(r_k)$ are pairwise disjoint. The same holds for $f(c_1), \dots, f(c_k)$. In contrast, each row and column intersect in exactly one coordinate. Their defect is

$$\|r_i + c_j\|_p^p + \|r_i - c_j\|_p^p - 2\|r_i\|_p^p - 2\|c_j\|_p^p = 2^p - 4 \neq 0.$$

Consequently $\text{supp}(f(r_i))$ meets $\text{supp}(f(c_j))$ for every (i, j) . A single coordinate can belong to at most one row support and at most one column support, so it can account for at most one of these k^2 intersections. It follows that $d \geq k^2$.

For $n \geq 9$, take $k = \lfloor (n - 1)/4 \rfloor$ and add distinct points if necessary to obtain an n -point set. Any exact embedding of that set restricts to one of $\mathcal{X}_k$, proving the lower bound in (1.3). The upper bound is Lemma 2.1. Finally, $0 < \gamma(p) < 1$, so the logarithmic limits in Theorem 1.1 follow from the stated bounds.

A Electrical-flow localization

The following proof follows the heat-kernel and entropy argument of Gurel-Gurevich, Nachmias, and Sachdeva [7, Sections 2–4], in the symmetric projection notation used here.

Proof of Lemma 4.1. Fix a strictly positive edge vector w , and define a vertex probability measure and its diagonal matrix by

$$\mu_i = \frac{\sum_{e \ni i} w_e^2}{2\|w\|_2^2}, \quad M = \text{diag}(\mu_i).$$

Connectivity implies $\mu_i > 0$. Put

$$L = B^T C B, \quad S = M^{-1/2} L M^{-1/2}, \quad K_t = M^{-1/2} e^{-tS} M^{-1/2}.$$

Writing $A = C^{1/2} B M^{-1/2}$, we have $S = A^T A$ and $\text{im } A = \text{im}(C^{1/2} B)$. Spectral decomposition therefore gives the convergent matrix integral

$$Q = \int_0^\infty C^{1/2} B K_t B^T C^{1/2} dt. \quad (\text{A.1})$$

Indeed, the integrand is $A e^{-tA^T A} A^T$, whose integral is the orthogonal projection onto $\text{im } A$.

For a vertex v , set $h_s = K_s \mathbf{e}_v$, where $\mathbf{e}_v$ is the v -th coordinate vector. The identity

$$K_s = e^{-sM^{-1}L} M^{-1}$$

shows that h_s has strictly positive coordinates for $s > 0$: the matrix $-M^{-1}L$ has nonnegative off-diagonal entries, and after adding a sufficiently large multiple of the identity, its exponential is strictly positive by connectivity and the power series for the exponential. Moreover,

$$\mu^T h_s = 1, \quad h_s \longrightarrow \mathbf{1} \quad (s \rightarrow \infty), \quad h_0 = \mu_v^{-1} \mathbf{e}_v.$$

Mass conservation follows from $\mathbf{1}^T L = 0$; the limit follows from the spectral decomposition of S , whose kernel is spanned by $M^{1/2} \mathbf{1}$, a unit vector.

For a positive vertex vector h , define

$$\mathcal{I}(h) = \sum_{e=ij} c_e (h(i) - h(j)) (\log h(i) - \log h(j)).$$

Using $\partial_s h_s = -M^{-1} L h_s$ and mass conservation, we obtain

$$\frac{d}{ds} \sum_i \mu_i h_s(i) \log h_s(i) = -h_s^T L \log h_s = -\mathcal{I}(h_s).$$

Integrating first over a compact subinterval of $(0, \infty)$ and then taking the endpoints to 0 and ∞ gives

$$\int_0^\infty \mathcal{I}(h_s) ds = \log(1/\mu_v). \quad (\text{A.2})$$

Here $0 \log 0 = 0$, so continuity gives the entropy at time 0 as $\log(1/\mu_v)$, whereas its limit at infinity is 0.

For $a, b > 0$, with the continuous interpretation when $a = b$,

$$\frac{a - b}{\log a - \log b} = \int_0^1 a^u b^{1-u} du \leq \frac{a + b}{2}.$$

Cauchy-Schwarz and $\mu^T h_s = 1$ consequently yield

$$\begin{aligned} (w^T C^{1/2} |B h_s|)^2 &\leq \mathcal{I}(h_s) \sum_{e=ij} w_e^2 \frac{h_s(i) + h_s(j)}{2} \\ &= \mathcal{I}(h_s) \|w\|_2^2. \end{aligned} \quad (\text{A.3})$$

Since $K_{2s} = K_s M K_s$, substituting $t = 2s$ in (A.1), expanding $K_s M K_s$ into column outer products, and applying the triangle inequality gives

$$\begin{aligned} w^T |Q| w &\leq 2 \sum_v \mu_v \int_0^\infty (w^T C^{1/2} |B K_s \mathbf{e}_v|)^2 ds \\ &\leq 2 \|w\|_2^2 \sum_v \mu_v \log(1/\mu_v) \leq 2 \|w\|_2^2 \log n, \end{aligned}$$

by (A.2) and (A.3). Continuity extends this bound to nonnegative w . Finally, $|Q|$ is symmetric and entrywise nonnegative, so for every real vector x ,

$$|x^T |Q| x| \leq |x^T |Q| |x| \leq 2 \log n \|x\|_2^2.$$

This proves (4.1). $\square$

B The Poisson moment inequality

Proof of Lemma 7.1. We prove the moment bounds first for a sum of independent symmetric real random variables $X_1, \dots, X_N$ with finite q th moments. For real a, b ,

$$\frac{|a+b|^q + |a-b|^q}{2} \geq |a|^q + |b|^q. \quad (\text{B.1})$$

Indeed, convexity of $u \mapsto u^{q/2}$ first bounds the left side below by $(a^2 + b^2)^{q/2}$, which is at least $|a|^q + |b|^q$. Iterating (B.1) proves the lower moment bound.

For every $\epsilon' > 0$ there is $C_{q,\epsilon'}$ such that

$$\frac{|a+b|^q + |a-b|^q}{2} \leq |a|^q + (1 + \epsilon')|b|^q + C_{q,\epsilon'} |a|^{q-2} |b|^2. \quad (\text{B.2})$$

To see this, the case $a = 0$ is immediate. Otherwise divide by $|a|^q$ and put $u = b/a$. Near $u = 0$, Taylor's theorem bounds the left side minus 1 by $C_q u^2$. For sufficiently large $|u|$, the left side divided by $|u|^q$ is at most $1 + \epsilon'$. On the remaining compact set, enlarge the coefficient of u^2 .

Let $S_j = X_1 + \dots + X_j$ and $M_q = \mathbb{E}|S_N|^q$. Each S_j is the conditional expectation of S_N given $X_1, \dots, X_j$, so $\mathbb{E}|S_j|^q \leq M_q$. Independence, symmetry, and (B.2) give

$$M_q \leq (1 + \epsilon') \sum_{j=1}^N \mathbb{E}|X_j|^q + C_{q,\epsilon'} M_q^{1-2/q} \sum_{j=1}^N \mathbb{E}|X_j|^2.$$

Young's inequality bounds the last term by $\theta M_q + C_{q,\epsilon',\theta} (\sum_j \mathbb{E}|X_j|^2)^{q/2}$. Choose $\epsilon', \theta > 0$ so that $(1 + \epsilon')/(1 - \theta) \leq 1 + \epsilon$, and absorb θM_q . This proves the asserted independent-sum bounds.

To pass to a Poisson sum on a region of mass $M > 0$, take $N > M$ independent trials, each equal to an independent signed sample with probability M/N and zero otherwise. The sums of their q th and second moments are exactly $\int |h|^q d\nu$ and $\int h^2 d\nu$. Their activation count converges in law to a Poisson variable of mean M . Conditional on that count, their nonzero samples have the desired law. If B_N is the activation count, then

$$\mathbb{E} e^{B_N} = \left(1 + \frac{M(e-1)}{N}\right)^N \leq e^{M(e-1)}.$$

Since h is bounded, these exponential bounds give uniform integrability of the q th powers of the sums. Weak convergence followed by truncation therefore gives convergence of their q th moments; see also [5, Sections 3.2 and 4.6] for these standard convergence principles. The independent-sum bounds therefore imply (7.9). The case $M = 0$ is immediate. $\square$

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